TAKE-HOME EXERCISE 2

Async client prioritization

Parachute in and improve: you inherit an existing system, diagnose it, and decide the right next thing to build. This document contains everything you need — the pre-read, the exercise brief, and the submission criteria.

PART ONE	PART TWO	PART THREE
Pre-read	Exercise brief	Final output

PART ONE

Pre-read

What you will want to know before running through the Async Client Prioritization exercise.

Before you start

1. You should have received a separate **candidate packet** covering Loom and dictation software setup. Go through it first so you can record and work comfortably alongside an agent.
2. Read the **Final Output Document** next, so you know exactly what a complete submission looks like.
3. Then read the **Exercise Brief**, including the pre-read packet inside it, and begin.

WORTH KNOWING

You will inherit an existing system and decide the right next thing to build. Budget 15–20 minutes for the pre-read before the 50–60 minute working session.

PART TWO

Exercise brief

A right-next-thing systems design exercise. It evaluates how you build a mental model quickly, tie technical decisions to product and business outcomes, prioritize under constraints, adapt when requirements change, and use an agent as a collaborator without outsourcing judgment.

Pre-read (async)	15–20 min	Interactive session	50–60 min
Artifacts	Short markdown docs, optional diagram	Recording	One Loom across the whole exercise, plus chat histories

RECORDING

Please create a single Loom recording of yourself running through the entire exercise. We want to hear you think out loud — treat it like pair designing.

Tools

ALLOWED

- Any AI assistant for quickly ramping on the pre-read packet
- Formatting your responses

NOT ALLOWED

- Outsourcing judgment — asking the model to solve the problem for you, in whole or in part

Pre-read packet

1 · Business brief

Client: **LatticeHQ** — mid-market B2B SaaS, roughly 1,200 customer orgs, SOC 2. Users are support agents plus self-serve customers via an in-app widget.

Support volume	Up 35% QoQ; backlog growing	Trust	Fragile — a few hallucinations caused escalations
Enterprise concern	Sensitive to tenant data leakage	North Star	Ticket deflection 8% → 18% within a 6-week pilot
Median time-to-first-response	4h → 1h	Hallucination complaint rate	Under 0.5% of copilot sessions
p95 latency	Under 1.2s	Infra cost	LLM + retrieval under $12k/month during pilot

CONSTRAINTS

- 6-week pilot; team of 2 engineers (backend + full-stack) and 1 PM
- SOC 2, with strict tenant isolation expectations
- No new vendor procurement during the pilot
- Safe refusal preferred over confident wrong answers
- Must ship something measurable in week 2 — a trust-building quick win

DECISION REQUIRED

What is the right next step that moves outcomes fast without dead-ending the longer-term path?

2 · System overview, current state

1. User asks a question (widget or agent console) with a `tenant_id`
2. `copilot-api` authenticates and resolves the tenant
3. Retriever queries the doc index and optional ticket summaries
4. LLM generates an answer with citations
5. Feedback stored — thumbs up/down plus an optional note

DATA SOURCES

- Knowledge base: Confluence pages, scraped nightly
- Public docs site, scraped daily
- Zendesk tickets — last 20 per tenant, summaries generated offline

ISOLATION CONSTRAINTS

- Enterprise: no cross-tenant data access, even in embeddings or the index
- Redaction mode for regulated tenants — no ticket snippets allowed

3 · Architecture diagram, intentionally incomplete

```
flowchart LR
U[User / Agent] -->|question + tenant_id| API[copilot-api]
API --> AUTH[Auth + Tenant Resolver]
AUTH --> RAG[RAG Orchestrator]
RAG --> RET[Retriever]
RET --> IDX[(Vector Index)]
RET --> DOCS[Doc Store]
RET --> TIX[Ticket Summary Store]
RAG --> LLM[LLM Gateway]
LLM --> RSP[Response + Citations]
RSP --> API
API --> U
U -->|thumbs up/down| FB[Feedback API]
FB --> FDB[(Feedback DB)]
```

4 · Observed symptoms

- **Tenant leakage near-miss:** one response cited a KB article ID from another tenant. Root cause unclear.
- **Quality:** answers are plausible but not grounded; citations often irrelevant.
- **Latency:** p95 occasionally 3–5s during bursts.
- **Cost:** token usage higher than expected — large contexts suspected.
- **Observability:** logs exist but are not structured; hard to answer "why did we respond this way?"
- **Feedback loop:** thumbs-down captured but unused; no eval gating before releases.

Interactive session · 50–60 minutes

PHASE 1
5–10 min

Handoff summary

Demonstrate fast, accurate sensemaking and boundary recognition of the problem and the architecture.

Deliverable: HANDOFF.md. You may use an agent to format the response — we recommend dictation — but the synthesis must be your own.

You may ask **one** clarifying question total during the session.

PHASE 2
20–25 min

Right next thing

Propose the next shippable milestone (B) tied to business goals, with a coherent path to Z.

Deliverable: PLAN.md. Optional: an updated Mermaid diagram.

PHASE 3
15–20 min

Curveballs

Adaptation, sequencing, and what you say no to. Respond to both; choose which to handle deeply first.

- **A · Trust and enterprise:** a large enterprise prospect will proceed only if you can credibly guarantee no cross-tenant leakage and provide auditability of sources used per answer — within the pilot.
- **B · Cost:** the CFO demands a 30% reduction in inference cost within 2 weeks without materially harming deflection.

Deliverable: CURVEBALLS.md.

PHASE 4
3–5 min

Decision log

A concise reflection, added as a section at the bottom of PLAN.md.

Templates

These are baselines to help you structure your response. If you have a better way to structure your thinking, please use it.

HANDOFF.MD · 300–600 WORDS

Come up with your own structure — one that adequately conveys the problem, the architecture, and the boundaries as you understand them.

PLAN.MD · RECOMMENDED 1–2 PAGES

- **A — current state:** 3–6 bullets on what is broken and what matters most.
- **B — next 2-week milestone:** what ships in 2 weeks that measurably moves the North Star, and why it is the best next move given constraints.
- **Z — 6–12 month direction:** what the mature system looks like, and how B stays on the upgrade path.
- **Proposal:** architecture delta — what you would add or change, optionally as an updated Mermaid diagram or component-change bullets.
- **Sequencing:** week 1, week 2, optionally weeks 3–6.
- **Risks and mitigations:** top five, each as risk → mitigation.
- **Validation plan:** success metrics — North Star plus leading indicators — evaluation and monitoring, and rollout.
- **Decision log:** what you said no to and why, the biggest tradeoff you made, what you would validate first, and what you would do with one more week.

CURVEBALLS.MD

For each curveball — trust and enterprise, then cost reduction — cover:

- What changes in my plan
- What I explicitly do not do now
- How I'd prove it

PART THREE

Final output

The submission criteria for the Async Client Prioritization exercise.

Loom recording

Please submit a single Loom screen recording that includes:

- Audio and video — at least audio; camera optional
- Your screen where appropriate

If your Loom gets cut off for any reason, multiple Looms are fine — but one is preferred.

BEFORE YOU SUBMIT

You should have received a separate packet detailing how to submit Loom videos. Follow those instructions so each recording is uploaded and shared to the right location in your private interview submission space.

Design artifacts

MARKDOWN DOCS	DIAGRAMS	CHAT HISTORIES
• HANDOFF.md • PLAN.md • CURVEBALLS.md	We encourage optional Mermaid diagrams where they help.	We encourage sharing private links to the chat histories used to create the documents above.